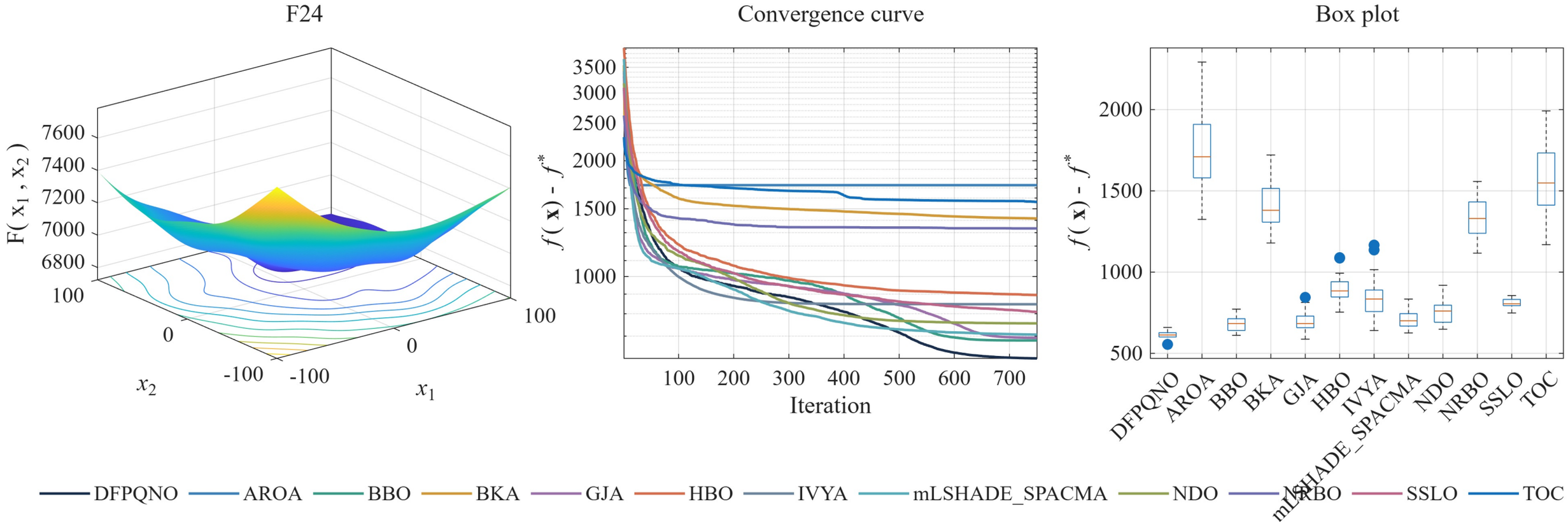

F24
F(x1 , x2)
7600
7400
7200
7000
6800
100
0
-100
x2
-100
0
100
x1
Convergence curve
f(x) - f*
3500
3000
2500
2000
1500
1000
100
200
300
400
500
600
700
Iteration
Box plot
f(x) - f*
2000
1500
1000
500
DFPQNO
AROA
BBO
BKA
GJA
HBO
IVYA
mLSHADE_SPACMA
NDO
NRBO
SSLO
TOC
DFPQNO
AROA
BBO
BKA
GJA
HBO
IVYA
mLSHADE_SPACMA
NDO
NRBO
SSLO
TOC